PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY INTERNSHIP | NETWORKWALKS

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Program /Batch	Cybersecurity/B084E
Date	11/NOV/26
Module Completed	1- W2-PM3: Maltego based Footprinting Attacks 2- W2-PM5: Zenmap based Network Scanning
Target	1- Networkwalks.com 2- My own Local LAN
Modules	1- Footprinting 2- Scanning

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using maltego software or tools (W2-PM3) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.
All the process are completed using the maltego software (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the

result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Maltego	OSINT and visual footprinting
Nmap	Scanning engine used by Zenmap
Windows CMD	Local IP and MAC address identification
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.

4. Activities Performed

4.1 Footprinting & Reconnaissance

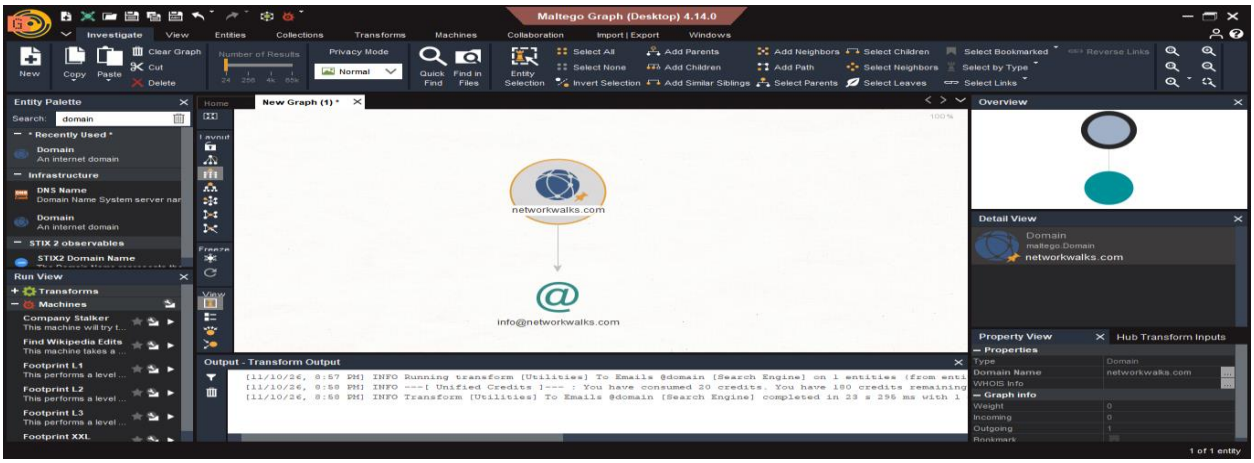
For the first activity, I used Maltego to perform OSINT-based footprinting against the authorized Networkwalks domain.

I started by opening Maltego and creating a new investigation graph. I added the target domain as the starting entity and explored the available transforms to collect publicly available information.

The purpose of this activity was to understand the relationships between domain information, DNS records, IP addresses and other publicly available infrastructure entities.

Methodology:

1. Opened Maltego and created a new graph.
2. Added the authorized target domain as the starting entity.
3. Selected and executed available transforms.
4. Observed the entities and relationships returned by Maltego.
5. Analyzed the generated graph.
6. Saved screenshots as practical evidence.



4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

Command Prompt

```
Microsoft Windows [Version 10.0.19045.7807]
(c) Microsoft Corporation. All rights reserved.

C:\Users\xpert computers>ifconfig
'ifconfig' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\xpert computers>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::8381:2fd7:95cc:8cf0%6
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

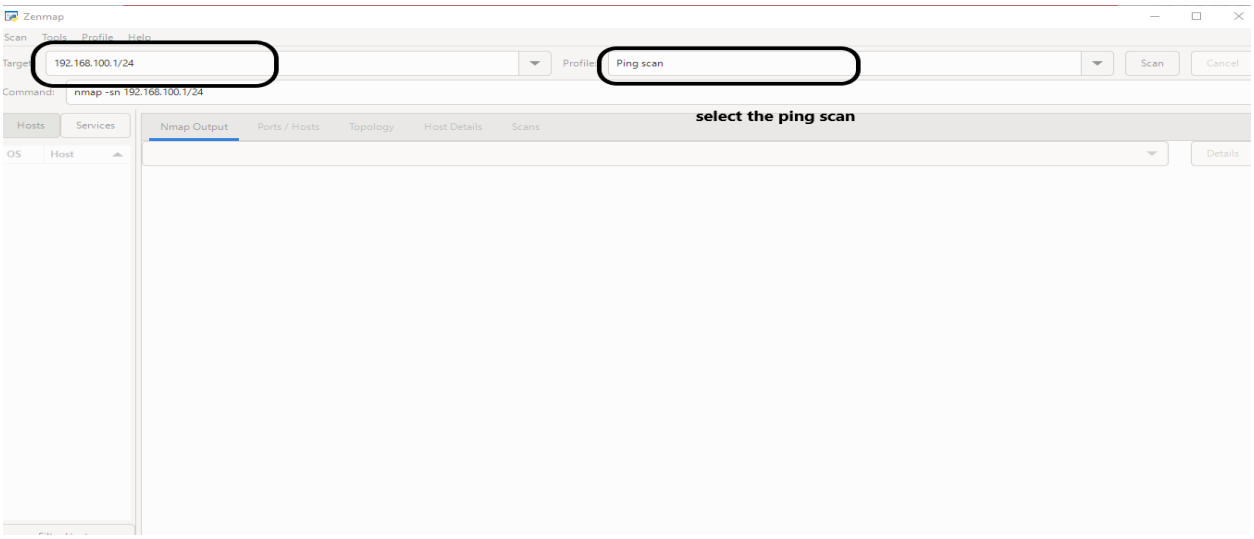
Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

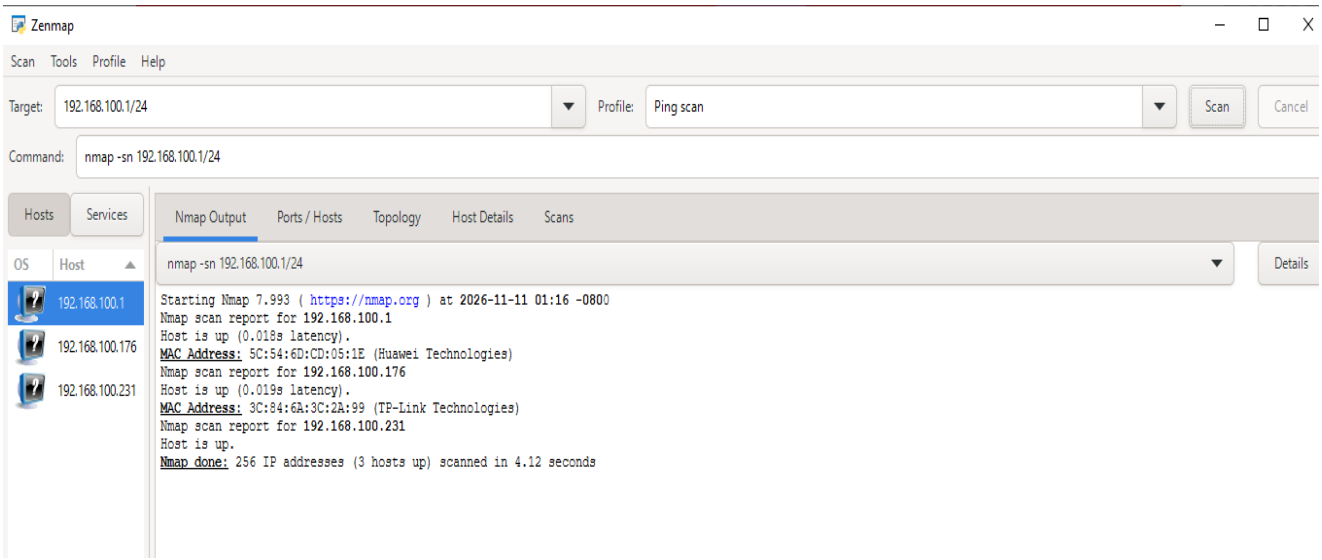
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::ddfe:8c91:a9ff:7c7a%5
   IPv4 Address. . . . . . . . . . . : 192.168.100.231
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%5
                                       192.168.100.1
```

Then open the nmap ,



Then click the scan button

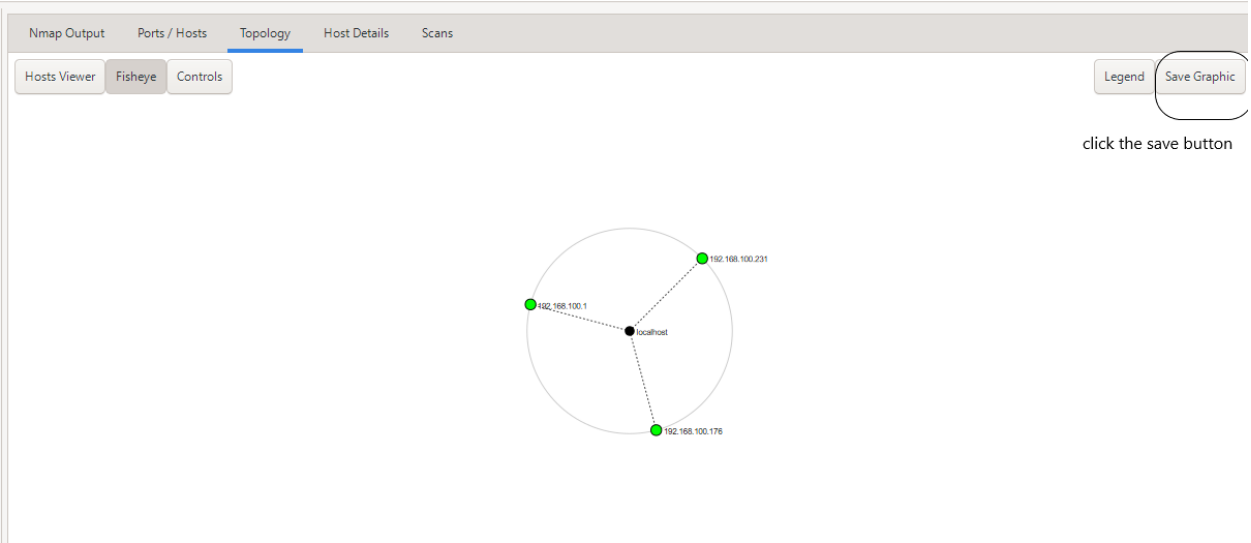


The results provided in the practical identified three live hosts:

- `192.168.100.1`
- `192.168.100.175`
- `192.168.100.231`

The results also included four MAC addresses.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.



Note: The actual subnet, number of hosts and addresses should be replaced with the results from my own network when submitting the report.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Public domain information exposed	Maltego identified publicly available domain-related information	May help attackers understand the organization's external digital footprint	🟡 **Low**
2	IP addresses associated with domain identified	Maltego displayed associated IP address entities (if observed)	Provides information about the hosting or network infrastructure	🟡 **Low**
3	Related infrastructure entities discovered	Maltego graph displayed relationships between discovered entities	May assist further infrastructure mapping	🟡 **Low**
4	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	🟠 **Medium**
5	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	🟠 **Medium**

Risk level key: 🔴 Critical 🟠 Medium 🟡 Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the findings, I recommend the following security improvements:

1. Review Public Information: Regularly monitor publicly available domain and infrastructure information.
2. Secure DNS Records: Review DNS records and remove unnecessary entries.
3. Monitor Network Devices: Perform regular network scans to identify active devices.
4. Investigate Unknown Devices: Verify unexpected devices connected to the network.
5. Maintain Network Documentation: Keep network topology and device records updated.
6. Authorized Testing: Perform security assessments only with proper permission.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship at Networkwalks, I completed two practical modules: Maltego-based Footprinting and Zenmap-based Network Scanning.

Using Maltego, I learned how to collect and visualize publicly available information through OSINT techniques. With Zenmap, I learned how to discover active hosts, identify network information and visualize network topology.

These activities improved my understanding of reconnaissance, footprinting and network discovery. I also learned the importance of documenting findings and performing security testing.